

LECTURE NOTES MATH255

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1. LECTURE 18

In this class we introduce the notion of a number field, these fields will be very important in the remainder as we will mainly study Galois theory in the context of number fields. We also introduce the notion of a multiplicative norm, and prove that certain subrings of number fields have a multiplicative norm.

1.1. Number fields. A number field, by definition, is a finite extension of the rational numbers. We will use the terminology (which is commonly used in number theory), that we denote number fields by the letter K .

Definition 1.1. *A number field K is a finite extension field of the rational numbers $\mathbb{Q}$.*

Example 1.1. Many of the examples we have covered in this class form an example of a number field. For example, $\mathbb{Q}(\sqrt{7})$ is a finite extension of $\mathbb{Q}$ and is therefore a number field. More generally, if p is any prime then the field $\mathbb{Q}(\sqrt{p})$ is a number field. In fact, if $\alpha \in \mathbb{C}$ is any algebraic number then $\mathbb{Q}(\alpha)$ is a finite extension of $\mathbb{Q}$ so $\mathbb{Q}(\alpha)$ is a number field¹. By picking multiple algebraic numbers $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ we can also form a finite extension as $\mathbb{Q}(\alpha_1, \dots, \alpha_n)$ so this is also an example of a number field.

Lemma 1.1. *Any number field K is an algebraic extension of $\mathbb{Q}$.*

Proof. Recall that any finite extension E of a field F is an algebraic extension of F . Applying this with $E = K$ and $F = \mathbb{Q}$ the lemma follows. ■

1.2. Multiplicative norms. Let R be an integral domain. We now introduce the notion of a multiplicative norm in R .

Definition 1.2. *A multiplicative $N : R \rightarrow \mathbb{Z}$ norm is a map with the following two properties:*

- (i) $N(a) = 0$ if and only if $a = 0$,
- (ii) $N(ab) = N(a)N(b)$ for all $a, b \in R$.

Example 1.2. The Euclidean norm on the Gaussian integers is an example of a multiplicative norm, we will see a far-reaching generalization of this.

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¹It is a non-trivial fact that the converse of this also holds, if K is a number field then there is some algebraic θ such that $K = \mathbb{Q}(\theta)$

Theorem 1.1. *Let R be an integral domain with a multiplicative norm N . Then $N(1_R) = 1$, and $|N(u)| = 1$ for all units $u \in R$. Moreover, if every $a \in R$ that satisfies $|N(a)| = 1$ is a unit, then if $a \in R$ satisfies that $|N(a)| = p$ is a prime number then $a \in R$ is irreducible.*

Proof. We begin by noting that $N(1_R) = N(1_R \cdot 1_R) = N(1_R) \cdot N(1_R)$ which implies that $N(1_R) = 1$ (recall that $N(1_R) \neq 0$ since this would imply that $1_R = 0_R$ but in an integral domain we have $1_R \neq 0_R$). If $u \in R$ is a unit then we have:

$$(1.1) \quad 1 = N(1_R) = N(uu^{-1}) = N(u)N(u^{-1})$$

which implies that $N(u) = \pm 1$ (since otherwise we would have $|N(u)N(u^{-1})| > 1$ recalling that both $N(u)$ and $N(u^{-1})$ are integers). Assume now that $u \in R$ is a unit if and only if $|N(u)| = 1$. Assume that $a \in R$ satisfies $|N(a)| = p$ is a prime. If we can write $a = bc$ then we have $p = |N(a)| = |N(bc)| = |N(b)| \cdot |N(c)|$ where both $|N(b)|$ and $|N(c)|$ are integers. Since p is prime, it follows that either $|N(b)| = 1$ or $|N(c)| = 1$. That is, either b or c must be a unit in R . Since in any decomposition $a = bc$ we deduce that one of the factors b or c is a unit, it follows that a is irreducible. ■

1.3. Multiplicative norms in number fields. Let K be a number field. That is, K is a finite extension of $\mathbb{Q}$. We can view K as a vector space over $\mathbb{Q}$ with some dimension $n = [K : \mathbb{Q}]$. If $\alpha \in K$ then we can define a linear map $A_\alpha : K \rightarrow K$ by:

$$(1.2) \quad A_\alpha(\beta) = \alpha\beta,$$

to see that A_α is a linear map we simply note that:

$$(1.3) \quad A_\alpha(\beta + c\gamma) = \alpha(\beta + c\gamma) = \alpha\beta + c(\alpha\gamma) = A_\alpha(\beta) + cA_\alpha(\gamma), \quad \beta, \gamma \in K, \quad c \in \mathbb{Q}.$$

Since A_α is a linear map on a finite dimensional vector space (over $\mathbb{Q}$) we can define a map $N : K \rightarrow \mathbb{Q}$ as follows:

$$(1.4) \quad N(\alpha) = \det(A_\alpha)$$

where $\det(A_\alpha)$ is the determinant of the linear map A_α .

Lemma 1.2. *The map N satisfies $N(\alpha\beta) = N(\alpha)N(\beta)$ and $N(\alpha) = 0$ if and only if $\alpha = 0$.*

Proof. The key observation is that the following holds: $A_{\alpha\beta}(\gamma) = \alpha\beta\gamma = \alpha(\beta\gamma) = A_\alpha(\beta\gamma) = A_\alpha(A_\beta(\gamma))$ so $A_{\alpha\beta} = A_\alpha A_\beta$. Using this in combination with properties of the determinant we have:

$$N(\alpha\beta) = \det(A_{\alpha\beta}) = \det(A_\alpha A_\beta) = \det(A_\alpha) \det(A_\beta) = N(\alpha)N(\beta),$$

which proves the first part of the lemma. To see the second part, note first that $A_0(\gamma) = 0 \cdot \gamma = 0$ so A_0 is the zero map and consequently $N(0) = \det(A_0) = 0$. For the second part of the lemma, note that $A_1(\gamma) = 1 \cdot \gamma = \gamma$ so $A_1 = \text{id}_K$ and therefore $N(1) = 1$. For any $\alpha \in K \setminus \{0\}$ we have $1/\alpha \in K$, since K is a field, so:

$$(1.5) \quad 1 = N(1) = N(\alpha/\alpha) = N(\alpha)N(1/\alpha)$$

from which it follows that $N(\alpha) \neq 0$. ■

If we fix a basis $e_1, \dots, e_n \in K$ over $\mathbb{Q}$ then we can write any A_α as a matrix with respect to this basis (explicitly we find the entries $a_{ij} \in \mathbb{Q}$ of the matrix as $A_\alpha(e_i) = a_{i1}e_1 + a_{i2}e_2 + \dots + a_{in}e_n$). It follows that we can define a map $\psi : K \rightarrow M_{n \times n}(\mathbb{Q})$ by:

$$(1.6) \quad \psi(\alpha) = A_\alpha$$

where A_α is understood as a matrix with respect to the basis $e_1, \dots, e_n$.

Lemma 1.3. *The map $\psi : K \rightarrow M_{n \times n}(\mathbb{Q})$ is a homomorphism of rings, and if we let $M(K) = \text{Im}(\psi)$ be the image of ψ then $\psi : K \rightarrow M(K)$ is an isomorphism (and in particular, $M(K)$ is a field).*

Proof. We have already shown that $\psi(\alpha\beta) = A_{\alpha\beta} = A_\alpha A_\beta = \psi(\alpha)\psi(\beta)$, and one can similarly prove that $\psi(\alpha + \beta) = \psi(\alpha) + \psi(\beta)$. It follows that ψ is a homomorphism. To show that ψ is an isomorphism when we restrict the codomain to its image, we simply note that if $\alpha \neq 0$ then $N(\alpha) = \det(A_\alpha) = \det(\psi(\alpha)) \neq 0$ so $\psi(\alpha)$ is invertible and therefore not 0. That is, the kernel of ψ is trivial, so ψ is injective. But then the map $\psi : K \rightarrow M(K)$ is both injective and surjective and therefore bijective. ■

Now we can introduce a certain subring of K on which we will get a multiplicative norm N . Let $M(K; \mathbb{Z}) \subset M(K)$ be the subset where every matrix has integer entries. We also define $\mathcal{O} = \psi^{-1}(M(K; \mathbb{Z}))^2$.

Lemma 1.4. *The subset $M(K; \mathbb{Z}) \subset M(K)$ is a subring that contains unity, and is therefore an integral domain. Consequently, $\mathcal{O}$ is a subring of K that is an integral domain.*

Proof. That $M(K; \mathbb{Z})$ is a subring is immediate since sums and products of matrices with integer entries has integer entries. That $M(K; \mathbb{Z})$ contains identity follows since $\psi(1) = \text{id}_k$ has integer entries with respect to any basis (it is always the identity matrix). That the same properties holds for $\mathcal{O}$ follows since $\psi : \mathcal{O} \rightarrow M(K; \mathbb{Z})$ is an isomorphism. ■

Theorem 1.2. *The restriction $N : \mathcal{O} \rightarrow \mathbb{Z}$ is a multiplicative norm on $\mathcal{O}$.*

Proof. We already know that $N(\alpha\beta) = N(\alpha)N(\beta)$ and $N(\alpha) = 0$ if and only if $\alpha = 0$. It remains to show that N actually takes values in $\mathbb{Z}$. To see this, simply note that if $\alpha \in \mathcal{O}$ then $A_\alpha = \psi(\alpha) \in M(k; \mathbb{Z})$ has integer entries, so $N(\alpha) = \det(A_\alpha)$ is an integer (since the determinant of a matrix with integer entries is always an integer). ■

Example 1.3. The calculations done in this section are somewhat abstract and complicated, so let's try and figure out what the map N looks like in a special. Consider the number field $K = \mathbb{Q}(\sqrt{7})$. We begin by fixing the basis $e_1 = 1, e_2 = \sqrt{7}$ of K over $\mathbb{Q}$. For any $\alpha = a + b\sqrt{7} \in K$ we calculate:

$$(1.7) \quad A_\alpha(e_1) = (a + b\sqrt{7}) \cdot 1 = a + b\sqrt{7} = a \cdot e_1 + b \cdot e_2,$$

$$(1.8) \quad A_\alpha(e_2) = (a + b\sqrt{7}) \cdot \sqrt{7} = a\sqrt{7} + 7b = 7b \cdot e_1 + a \cdot e_2,$$

²This subring $\mathcal{O}$ is an example of an *order*, which is a certain subset of *algebraic integers* that are studied extensively in number theory.

so with respect to this basis we obtain the matrix:

$$(1.9) \quad \psi(\alpha) = \psi(a + b\sqrt{7}) = \begin{pmatrix} a & 7b \\ b & a \end{pmatrix}.$$

Recall that we define $\mathcal{O} \subset K$ by the property that $\psi(\alpha)$ should have integer entries. But:

$$(1.10) \quad \psi(a + b\sqrt{7}) = \begin{pmatrix} a & 7b \\ b & a \end{pmatrix}$$

have integers entries if and only if a and b are integers. It follows that in this case $\mathcal{O} = \{a + b\sqrt{7} \mid a, b \in \mathbb{Z}\} = \mathbb{Z}[\sqrt{7}]$. So we found the subring of $\mathbb{Q}(\sqrt{7})$ as $\mathbb{Z}[\sqrt{7}]$. Finally we calculate N as follows:

$$(1.11) \quad N(a + b\sqrt{7}) = \det(A_\alpha) = \det \begin{pmatrix} a & 7b \\ b & a \end{pmatrix} = a^2 - 7b^2.$$

That is, in the case $K = \mathbb{Q}(\sqrt{7})$ the calculations done in this section shows that $N(a + b\sqrt{7}) = a^2 - 7b^2$ is a multiplicative norm on $\mathbb{Z}[\sqrt{7}]$. Of course in this special case there are faster ways to show this, but the upside of the approach in this section is that it works for any number field K .

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